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Numerical investigation of measurement error of the integrating sphere based on the Monte-Carlo method



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HIGHLIGHTS

- Simulate the radiation transfer of multiple reflections inside integrating sphere using MCM.
- Investigate the effects of nonideal spherical structure of real integrating sphere.
- Correct the measured reflectance according to its simulation result.
- Propose the idea to optimize reflectance of coating and standard white plate.

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ABSTRACT

To accurately measure the directional-hemispherical reflectance of a material surface using an integrating sphere, we investigated the effects of the integrating sphere's interior structure and coating properties on the measured data by numerical simulation. The results showed that the integrating sphere's structure and the reflectance of the coating codetermine the accuracy of the measured results. When the integrating sphere's structure is fixed, choosing an appropriate internal coating using the method proposed in this paper ensures the accuracy of the results. If a proper coating is not available, we propose a method to effectively correct the measured results.

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1. Introduction

With the increasing demand for new materials in aerospace, new energy utilization, biomedicine, machining, and so on, processing and manufacturing of materials are rapidly developing [1–3]. However, utilization of new materials cannot be performed without accurate characterization of the materials' properties. As an important radiation property parameter, the accurate measurement of reflectance is extremely important [4–6]. There are various methods to measure the reflectance of materials [7–9], and the directional-hemispherical reflectance is frequently measured by an integrating sphere [10–13]. Investigation of the integrating sphere is also common. Symons [14] et al. designed a new apparatus to perform measurements of the solar transmittance of planar and nonplanar transparent samples for various angles of incidence. Dam [15] et al. presented a new method to extract the optical

properties from integrating sphere measurements of thin biological samples based on multivariate calibration techniques, multiple polynomial regression, and a Newton-Raphson algorithm for solving nonlinear equation systems. Bittar and Hamlin [16] designed a normal incidence absolute reflectometer based on an integrating sphere that allows accuracies of $\pm 0.2\%$ to be achieved over the 350–750 nm wavelength range. Prokhorov [17] et al. described a newly developed algorithm that substantially improves the convergence by calculating the direct source-induced irradiation for every point of the diffuse reflection of rays traced, and applied this to design an integrating sphere reflectometer for visible and infrared spectral ranges.

In an error study of the integrating sphere, Kim and Philpot [18] presented a new integrating sphere configuration for measuring back scattering using a semispherical cuvette, avoiding the error because of total internal reflection. Davies and Zagieboylo [19] proposed an improved integrating sphere that can determine the average reflectance and transmittance, using the low average reflectance of the coating (pure aluminum oxide) to reduce the error owing to non-uniform spectral response. Hanssen [20]

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Nomenclature

MCM	Monte-Carlo method
λ	wavelength
$i_{\lambda, \text{sample}}$	current value of corresponding detector for the sample at a wavelength of λ
$i_{\lambda, \text{white}}$	current value of corresponding detector for the standard white plate at a wavelength of λ
E_{in}	radiant energy of irradiation light entering the integrating sphere
E_{out}	radiant energy reaching the detector
fa	area of a certain open pore in the integrating sphere divided by the total area of its inner wall

fa_1	area of the detectors in the sphere divided by the total area of its inner wall
R_0	uniformly distributed random Number of Zenith Angle θ

Greek symbols

$\rho_{\lambda, \text{sample}}$	spectral reflectance of the sample at a wavelength of λ
$\rho_{\lambda, \text{white}}$	spectral reflectance of the standard white plate at a wavelength of λ
ρ	reflectance of the integrating sphere's inner wall
$\rho_{\lambda, \text{sample}}^*$	sample spectral reflectance at a wavelength of λ

investigated the error caused by non-Lambertian scattering of the interior wall of an integrating sphere through a model for measurement of directional-hemispherical reflectance. Park and co-workers [21–23] reported a series of studies on error correction of the integrating sphere. They realized differential spectral responsivity measurement using a light-emitting diode (LED)-based integrating sphere source and corrected the error caused by the lock-in detection spatial irradiance distribution and broad spectral bandwidth of the LEDs. In addition, they designed a six-port integrating sphere photometer and validated its capability of reducing systematic error because of spatial mismatch between the reference source and the source under test in the measurement of the total luminous flux.

When measuring the directional-hemispherical reflectance of solid materials, the light reflected from the sample surface in all directions needs to be concentrated on the photoelectric detector. One common method is realized by integrating spheres, which is also often used for measuring the lighting source. The practical performance of integrating spheres may not be desirable because of many effects and restrictions when it is used to accurately measure the sample spectral reflectance [24]. Thus, it is necessary to analyze the accuracy of the measurement, determine whether it is acceptable, and possibly correct the measured value. On the other hand, it is very difficult to simulate the measurement using the method of solving discrete equations. So random sampling by Monte-Carlo method is often used to achieve this, and relative investigation is also extensive now [25–26]. Li [27] et al. presented a backward Monte Carlo method based on radiation distribution factor to compute the apparent directional emissivity of one-dimensional absorbing-emitting-scattering semitransparent slab with specular semitransparent surface and opaque diffuse substrate. Parviainen and Muinonen [28] explored the distribution of the reflectance over the scattering hemisphere in particulate porous random media with macroscale surface roughness using Monte-Carlo ray tracing. Xiong and Shi [29] applied the polarized Monte Carlo ray tracing technique to the computer generated bicontinuous medium, so that the snow surface albedo, bidirectional reflectance distribution function can be simulated. Wan and Guo [30] did a numerical study for measuring cerebral blood oxygen saturation noninvasively using optical reflectance ratio of dual wavelengths, and used Monte Carlo method to simulate reflectance measurements in a model neck tissue.

In this study, a Monte-Carlo method (MCM) is used to perform a numerical simulation of the geometry construction of a typical integrating sphere system. According to simulations of different reflectance backgrounds, we first built an error correction model of the reflectance for this integrating sphere system. We then measured the surface hemispherical reflectance of various typical construction material samples, and effectively corrected the measured values using this model. Finally, based on comparison between the

corrected and measured results, we analyzed the relationship between the coating reflectance and the measurement error for this system, and determined the optimal integrating sphere coating properties.

2. Experimental system and measurements

The main part of the integrating sphere is a spherical cavity with a coating of high diffuse reflectance on its inner wall. There are several open pores in the wall to allow irradiation light in, place the material samples, place the photoelectric detector, and so on. The sample reflects the irradiation light, which is repeatedly reflected by the spherical cavity's inner wall. This produces diffuse light of uniform luminance in the inner wall, some of which enters the photoelectric detector.

Fig. 1 shows the experimental reflectance measurement system used in this study, and Fig. 2 shows a schematic diagram of its structure. The composite lamp composed of a deuterium lamp and a tungsten lamp can provide sustainable light with a continuous spectrum in the wavelength range 200–2500 nm. The wave chopper performs pulse modulation of the irradiation light, transforming the sustainable light into pulsed light, which is then converted by the filter wheel and the monochromator into monochromatic light with a narrow wavelength range. The monochromatic light reaches the standard white plate or sample wafer after entering the integrating sphere, and the light repeatedly reflected by the inner coating is transformed into a current signal by the detector. Finally, the current signal from the detector and the synchronizing signal from the chopper are exported to the lock-in amplifier, and the data are exported to the computer for post-processing.

Fig. 3 shows the size and structure of the integrating sphere in this system. The open pore at the bottom of the spherical cavity is the entrance port for the irradiation light, and the pore at the top is



Fig. 1. Experimental reflectance measurement system.

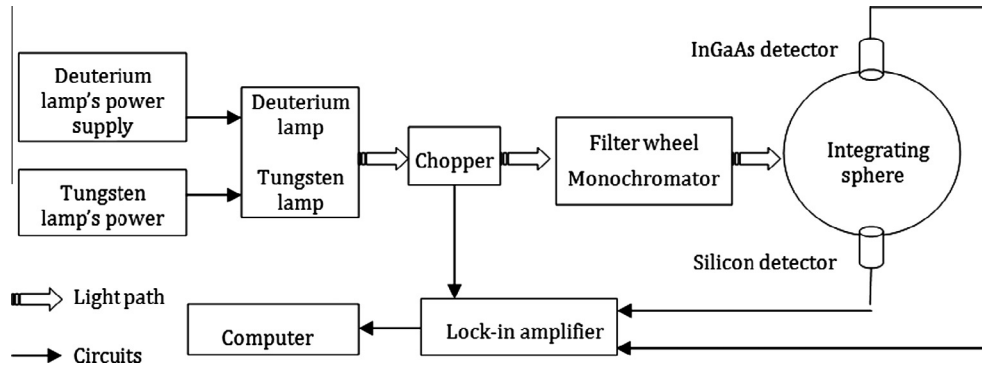


Fig. 2. Schematic diagram of the directional-hemispherical experimental reflectance measurement system.

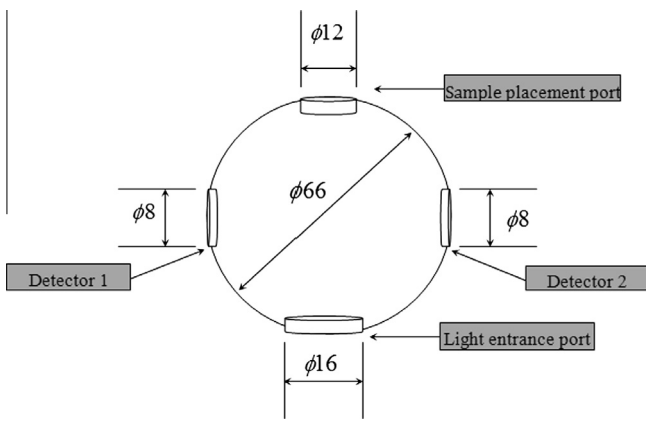


Fig. 3. Size and structure of the integrating sphere.

the placement port for the standard white plate or sample wafer. The symmetrical open pores on the left and right sides are used to place the silicon and InGaAs detectors.

This experimental system uses the contrastive measurement method to measure the reflectance of samples. First, the standard white plate is placed onto the sample port, then the required spectral band is scanned to obtain the current values of the corresponding detectors for each wavelength within the wave band. Second, the standard white plate is replaced with the sample to be tested, then the spectral band is scanned again to obtain the current values of the corresponding detectors for the sample at each wavelength. Finally, the measuring software in the computer multiplies the ratio of both current values at the same wavelength by the known spectral reflectance of the standard white plate to give the spectral reflectance of the sample:

$$\rho_{\lambda, \text{sample}} = \rho_{\lambda, \text{white}} \frac{i_{\lambda, \text{sample}}}{i_{\lambda, \text{white}}} \quad (1)$$

Eq. (1) is based on the hypothesis that the current of the detector (which is proportional to the light energy received by the detector) linearly changes with the spectral reflectance of the sample. However, the spherical structure of the integrating sphere is actually incomplete because of the several open pores in the inner wall. But actually, the spherical structure is not complete due to the several open pores in the integrating sphere's inner wall, and the surface properties of its inner wall is nonuniform owing to the sample and standard white plate's absorption of the light which is repeatedly reflected inside the integrating sphere. What's more, the reflectance of its coating under different wavelengths is fluctuant because of multiple reflection inside the integrating sphere. For

these reasons, measuring result is changing nonlinearly under different spectrums.

Fig. 4 shows the reflectance curve for an integrating sphere coating based on polytetrafluoroethylene (PTFE, commonly known as Teflon). The spectral reflectance varies between 0.93 and 0.99 in the wavelength range 300–2500 nm. It only slightly varies in the wavelength range 400–1900 nm, with a relatively large variation in the other wavelength ranges (300–400 nm and >1900 nm). As a consequence, the measurement error of the sample reflectance in these other wavelength ranges is enlarged.

When the integrating sphere is applied to measure irradiation light, based on the hypothesis that the spherical inner wall is complete and the sizes of the open pores are much less than the spherical diameter, a common equation to calculate the detector receiving efficiency is

$$\frac{E_{in}}{E_{out}} = \frac{\rho f a_1}{1 - \rho(1 - \sum f a)} \quad (2)$$

where E_{in} is the radiant energy of irradiation light entering the integrating sphere, E_{out} is the radiant energy reaching the detector, ρ is the reflectance of the integrating sphere's inner wall, $f a$ is the area of a certain open pore in the integrating sphere divided by the total area of its inner wall, and $f a_1$ is the area of the detectors in the sphere divided by the total area of its inner wall.

However, the area of the open pores occupies more than 10% of the total spherical area in this system, and there is a cylindrical 1-mm-thick cavity between the sample placement face and integrating sphere's inner wall, leading a noticeable error in the actual results compared with Eq. (2). In addition, the multiple scattering effect of light inside the integrating sphere caused by the sample or standard white plate is difficult to describe. Thus, it is necessary

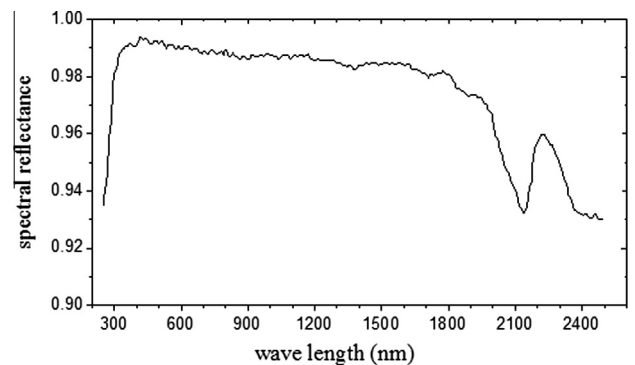


Fig. 4. Spectral reflectance of a typical PTFE coating material from FluorilonTM, (Avian Technologies).

to simulate radiation transfer inside the integrating sphere to verify the measurement accuracy and attempt to correct the measured results.

In this study, we performed a numerical simulation of radiation transfer inside the integrating sphere in this experimental system using a program developed by our group. The detector efficiencies with different spectral reflectances of integrating sphere coatings and with different sample reflectances can be obtained, which can then be used to correct the measured reflectance data.

3. Introduction of the Monte-Carlo method

Radiation transfer simulation using the MCM is usually a physical model based on geometrical optics, without including wave optics [31]. Even though some MCM simulations involve wave optics phenomena such as interference and diffraction, they only build random probability models for relevant phenomena using existing physical equations rather than simulating the mechanism of wave optics [32–34]. A MCM model based on geometrical optics was used in this paper, where the photon (light beam, light ray, ray and energy beam) transmission was divided into the following processes and the corresponding random model was built:

- (1) Surface (interface): emission, absorption, reflection, transmission and refraction.
- (2) Medium or medium with particles (dispersion): emission and attenuation (including absorption and scattering of medium or particles).

This is achieved by generating uniform random numbers R_0 in the interval $[0, 1]$, then deciding whether the physical phe-

nomenon occurs by comparing the magnitude of the random number R_0 and the formula result for describing the physical process, or calculating the physical phenomenon's condition according to R_0 . Because there are many relevant models and corresponding introductions, no more details will be given here.

4. Simulation conditions

The geometrical parameters used for the experiments are given in Fig. 3. The sample placement plane was a tangent to the integrating sphere's inner wall, and there was a 12-mm-diameter cylindrical surface between them. The material of the standard white plate was the same as that of the integrating sphere's internal coating, that is, they have the same spectral reflectance. The incident light was assumed to be parallel light with a diameter of 10 mm and irradiate to the sample placement plane vertically. It was also assumed that the detector surface can completely absorb light of all wavelengths.

Fig. 5 shows the efficiency of light reaching the detector using coatings with different spectral reflectances when the standard white plate was placed in the sample port. The efficiency change owing to different spectral reflectances is nonlinear, and the result of Eq. (2) is clearly different to the numerical simulation result.

5. Error correction model for experimental results

We performed simulations of the detector receiving efficiency when placing the standard white plate and samples with different reflectances in the sample port. During the calculations, the spectral reflectance of the coatings was between 0.90 and 0.99. The spectral reflectance of the samples was between 0.1 and 0.99.

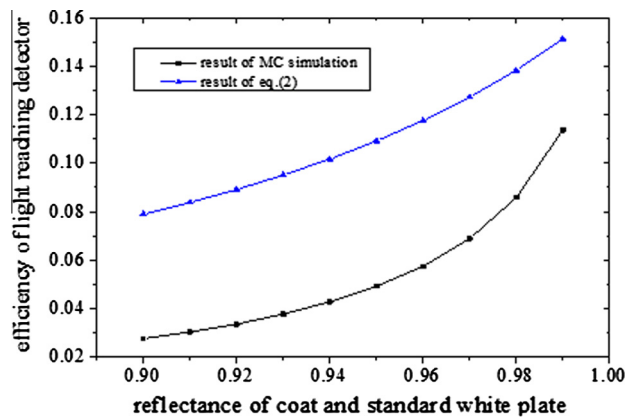


Fig. 5. Nonlinear relationship between coating reflectance and detection efficiency.

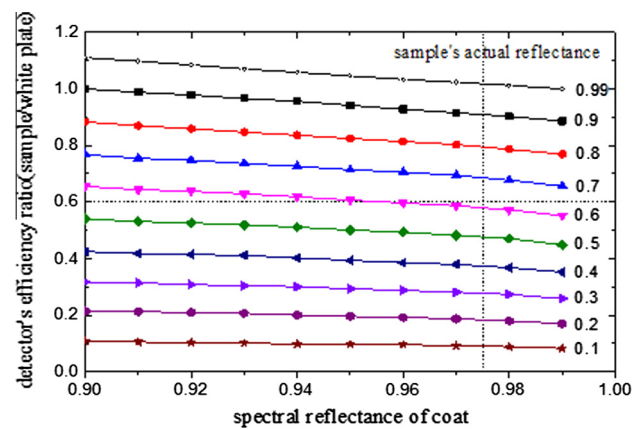


Fig. 6. Curves of detector efficiency ratio against spectral reflectance of the coating.

Table 1
Detector receiving efficiency of integrating sphere.

Sample reflectance	Spectral reflectance of coating and standard white plate									
	0.90	0.91	0.92	0.93	0.94	0.95	0.96	0.97	0.98	0.99
0.1	0.00296	0.00319	0.00345	0.0038	0.00421	0.00475	0.00549	0.00633	0.00761	0.00944
0.2	0.00588	0.00645	0.00702	0.00779	0.00859	0.00963	0.01103	0.01288	0.0154	0.01932
0.3	0.00876	0.00954	0.01034	0.01146	0.01288	0.01443	0.01654	0.01937	0.02347	0.02936
0.4	0.01173	0.01268	0.01394	0.0155	0.01726	0.01937	0.02217	0.02613	0.0316	0.04004
0.5	0.0149	0.01612	0.01769	0.01956	0.02189	0.02466	0.02837	0.0332	0.04041	0.05102
0.6	0.01809	0.01953	0.02144	0.02372	0.02649	0.02988	0.0343	0.04044	0.04912	0.0628
0.7	0.0212	0.0229	0.02513	0.02776	0.0311	0.03517	0.04057	0.04783	0.05813	0.07466
0.8	0.02439	0.02639	0.02887	0.03195	0.03582	0.04059	0.04679	0.05533	0.06754	0.08751
0.9	0.0276	0.02999	0.03288	0.03643	0.04096	0.04638	0.05334	0.06306	0.07748	0.10075
0.99	0.03067	0.0333	0.03645	0.04037	0.04534	0.05151	0.05941	0.07049	0.08691	0.1137
Standard white plate	0.0276	0.03033	0.0336	0.03771	0.04282	0.04924	0.05748	0.06887	0.08583	0.1137

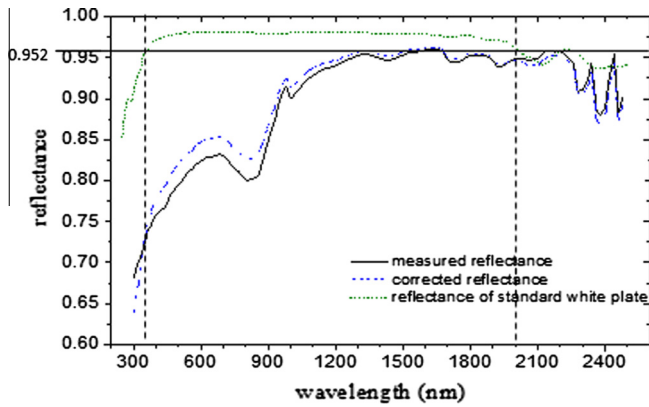


Fig. 7. Correction of the measured sample spectral reflectance.

Table 1 gives the simulated detector receiving efficiencies with coatings and samples with different spectral reflectances.

Because the experimental measurement scheme uses spectral reflectance of a standard white plate as a reference and the measuring software gives results based on Eq. (3), we obtained detector receiving efficiency ratio data of the samples and standard white plate for the spectral reflectance in Table 1. The curves of the data are shown in Fig. 6.

From Fig. 6, a distinct difference between the reflectance curve slopes of different samples can be observed. The slope of the curve is higher when the actual sample reflectance increases. The slope of each curve is not constant, but decreases with increasing spectral reflectance of the integrating sphere coating.

It is feasible to correct the sample spectral reflectance data obtained from this experimental system using the data in Fig. 6, and the method is as follows. Given the spectral reflectance of

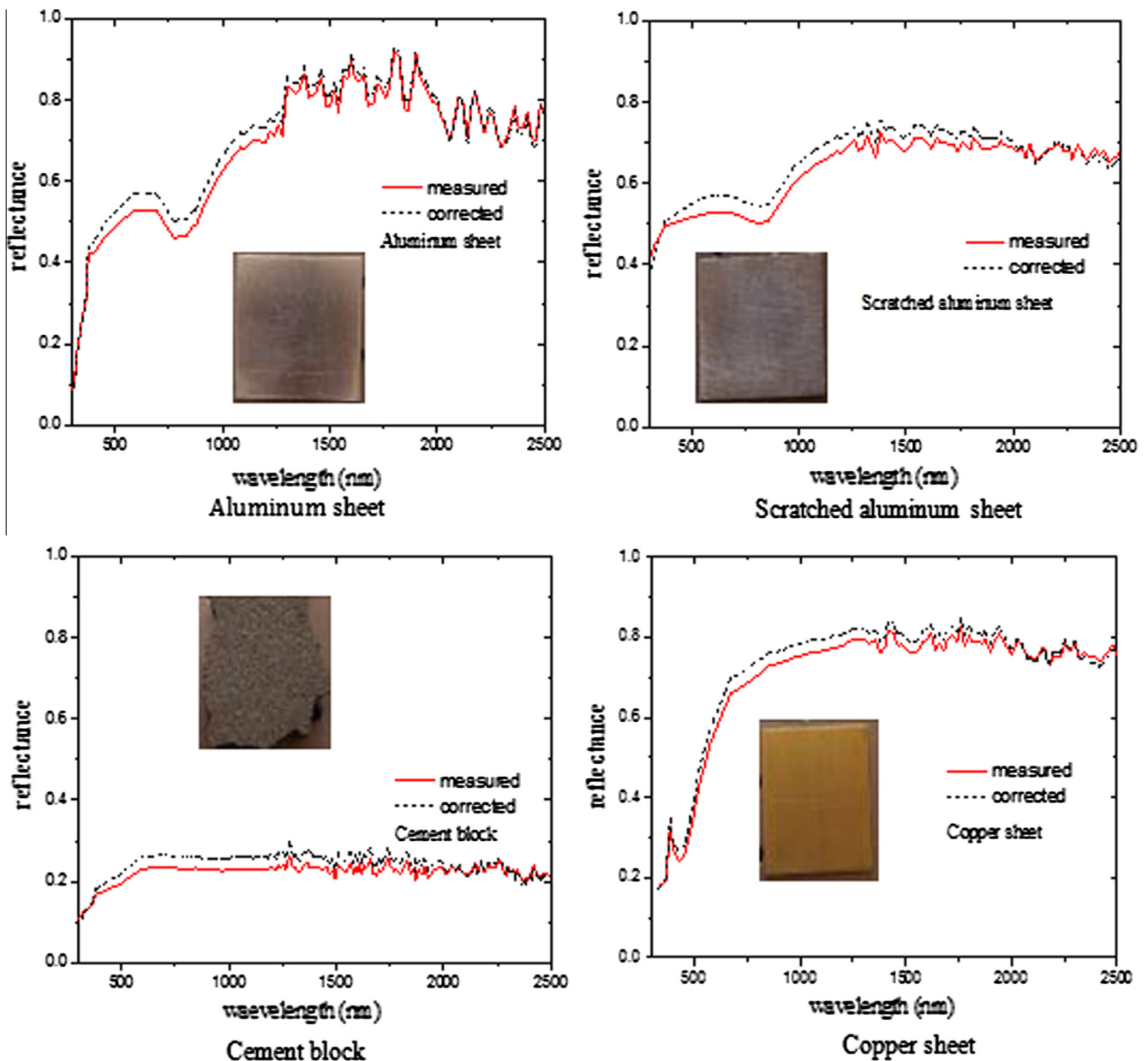


Fig. 8. Measured and corrected results for different surface material samples.

the integrating sphere coating and standard white plate within the measured spectral range, and the measured value of the sample spectral reflectance $\rho_{\lambda, \text{sample}}^*$ at a wavelength of λ , the spectral reflectance of standard white plate $\rho_{\lambda, \text{white}}$ at that wavelength can be determined. Because the detector current is proportional to the detector receiving efficiency when the irradiation source is stable, from Eq. (1)

$$\frac{\rho_{\lambda, \text{sample}}^*}{\rho_{\lambda, \text{white}}} = \frac{i_{\lambda, \text{sample}}}{i_{\lambda, \text{white}}} = \frac{\text{detector efficiency with sample}}{\text{detector efficiency with standard white plate}} \quad (3)$$

The numerical value from Eq. (3) corresponds to a certain vertical coordinate value in Fig. 6. Through this value, we draw a horizontal line, and through $\rho_{\lambda, \text{white}}$ on horizontal axis, we draw a vertical line. At the intersection point of the two lines, a data curve should pass through, the reflectance value which the data curve represents is the corrected sample spectral reflectance. For the situation shown by the dotted line in Fig. 6, the intersection point of the two dotted lines is not on any curves shown, we can interpolate the adjacent curves to obtain the corrected spectral reflectance, in this paper, we use the nearest four data points to make a linear interpolation.

Fig. 7 shows the spectral reflectance of a certain sample using this experimental system and the corrected reflectance according to the curves of spectral reflectance of the coating.

Actually, because the spectral reflectance of the integrating sphere coating and standard white plate is influenced by many factors, such as spraying technology, coating thickness, temperature, humidity, dust pollution, and long-time aging, a definite error would exist in the obtained data results if we used the same spectral reflectance curves of the coating and white plate under different experimental conditions.

6. Specific application

We measured the hemispherical transmittance of some typical surface materials. The hemispherical reflectances for an aluminum sheet, a scratched aluminum sheet, a cement block, and a copper sheet at wavelengths between 300 and 2500 nm are shown in Fig. 8.

After we obtained the experimental data, we used the developed integrating sphere correction model to correct the data, and

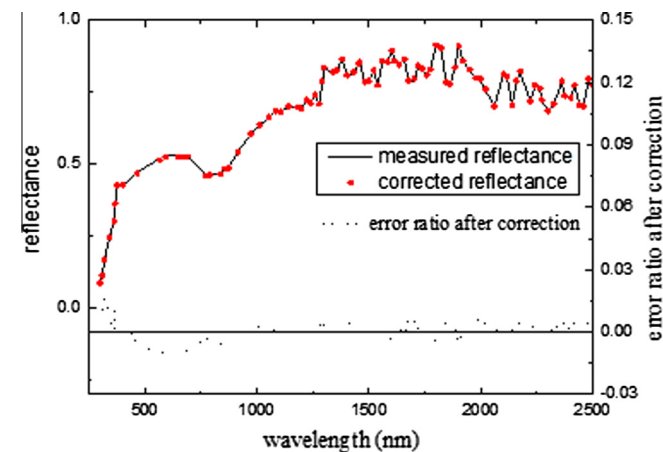


Fig. 9. Difference between measured and corrected data with a coating with reflectance 0.952.

the results are shown as dashed lines in Fig. 8. From the results, for the different materials considered, the measured reflectance was considerably different to the corrected reflectance in the visible and near-infrared regions (about 300–1900 nm), while there was little difference between the measured and corrected results at wavelengths between 1900 and 2500 nm.

According to the corrected results, we believe that the reason is because of the optical properties of the integrating sphere's coating. Combining with the error between measured and corrected reflectance showed in Fig. 7, with change of the coating reflectance in the direction of spectrum, the error between the sample reflectance and the corrected results changes. In the wavelength range between the two dashed lines in Fig. 7, the coating reflectance is greater than 0.952, and the corrected results are greater than the measured data. Outside this range, the coating reflectance is less than 0.952, and the corrected data are less than the measured data. Thus, it is speculated that for an integrating sphere with this structure, when the coating reflectance is 0.952, the corrected reflectance data are equivalent to the measured results. That is, if the coating reflectance is 0.952, we can consider that the measured data is accurate and does not require correction.

To validate this, we assumed the integrating sphere coating reflectance to be 0.952, then corrected the measured reflectance of the standard sample in Fig. 7, and the results are shown in Fig. 9. From the results, the corrected data basically agrees with the measured data, so we can consider that the measured results are accurate and do not require correction.

As a consequence, for the integrating sphere used in this study, a coating material with a reflectance of about 0.95 is required to obtain accurate measured results. Otherwise, we need to use the developed numerical correction method and the curves in Fig. 6 to correct the measured data.

7. Conclusions

In this paper, we performed numerical simulation of the measuring process of sample reflectance with an integrating sphere based on the MCM, and analyzed an integrating sphere with typical structure and coating. From the analysis, the measured reflectance of samples was different to the actual reflectance considering the integrating sphere's own structure and repeated interior scattering. In this case, we performed a numerical analysis of the error in the measured data. We then raised a numerical method to correct measured reflectance of samples, and constructed correction curves of the corresponding coating reflectance and measured reflectance. We then measured the reflectance of four typical samples and corrected the reflectance results. By analyzing the measured and corrected data, for an integrating sphere with this structure, when the coating reflectance is 0.952, the corrected reflectance data is equivalent to the measuring result, and thus the measuring data can be used directly without the need for correction. Otherwise, the measured results need to be corrected. Through these studies, we can improve the measurement precision of the integrating sphere measurement system and specifically choose an internal coating material, providing guidance for structural design as well as material choice of integrating spheres.

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References

- [1] F.H. Song, P.J. Yang, C.L. Xu, S.M. Wang, An improved global rainbow refractometry for spray droplets characterization based on five-point method and optimization process, *Flow Meas. Instrum.* 40 (2014) 223–231.
- [2] F.H. Song, C.L. Xu, S.M. Wang, Rainbow technique for multi-parameter measurement of absorbing cylinder, *Particuology* 11 (2013) 184–188.
- [3] X.C. Li, L.H. Li, J.M. Zhao, J.Y. Tan, Optical properties of sodium chloride solution within the spectral range from 300 to 2500 nm at room temperature, *Appl. Spectrosc.* 69 (2015) 635–640.
- [4] H. Qi, D.L. Wang, S.G. Wang, L.M. Ruan, Inverse transient radiation analysis in one-dimensional non-homogeneous participating slabs using particle swarm optimization algorithms, *J. Quant. Spectrosc. Radiat. Transfer* 112 (2011) 2507–2519.
- [5] R. Durikovic, A. Mihalik, Modeling the BRDF from spectral reflectance measurements of metallic surfaces, *Appl. Surf. Sci.* 312 (2014) 87–90.
- [6] B. Zhang, H. Qi, Y.T. Ren, S.C. Sun, L.M. Ruan, Inverse transient radiation analysis in one-dimensional participating slab using improved Ant Colony Optimization algorithms, *J. Quant. Spectrosc. Radiat. Transfer* 133 (2014) 351–363.
- [7] S. Nevas, F. Manoocheri, E. Ikonen, Gonireflectometer for measuring spectral diffuse reflectance, *Appl. Opt.* 43 (2004) 6391–6399.
- [8] W.H. Venable, J.J. Hsia, V.R. Weidner, Establishing a scale of directional-hemispherical reflectance factor I: the Van den Akker method, *J. Res. Natl. Bur. Stand.* 82 (1977) 29–55.
- [9] W. Budde, Calibration of reflectance standards, *J. Res. Natl. Bur. Stand. Sect. A* 80A (4) (1976) 585–595.
- [10] G.D. Vries, J.F. Beek, G.W. Lucassen, M.J.C.V. Gemert, Effects of light losses in double integrating spheres on optical properties estimation, *J. Sel. Top. Quant. Electron.* 5 (1999) 944–947.
- [11] E. Hawe, C. Fitzpatrick, P. Chambers, G. Dooly, E. Lewis, Hazardous gas detection using an integrating sphere as a multipass gas absorption cell, *Sens. Actuat. A* 141 (2008) 414–421.
- [12] D. Fukutomi, K. Ishii, K. Awazu, Highly accurate scattering spectra of strongly absorbing samples obtained using an integrating sphere system by considering the angular distribution of diffusely reflected light, *Lasers Med. Sci.* 30 (2015) 1335–1340.
- [13] K.A. Snail, L.M. Hanssen, Integrating sphere designs with isotropic throughput, *Appl. Opt.* 28 (1989) 1793–1799.
- [14] J.G. Symons, E.A. Christie, M.K. Peck, Integrating sphere for solar transmittance measurement of planar and nonplanar samples, *Appl. Opt.* 21 (1982) 2827–2832.
- [15] J.S. Dam, T. Dalgaard, P.E. Fabricius, S. Andersson-Engels, Multiple polynomial regression method for determination of biomedical optical properties from integrating sphere measurements, *Appl. Opt.* 39 (2000) 1202–1209.
- [16] A. Bittar, J.D. Hamlin, High-accuracy true normal-incidence absolute reflectometer, *Appl. Opt.* 23 (1984) 4054–4057.
- [17] A.V. Prokhorov, S.N. Mekhontsev, L.M. Hanssen, Monte Carlo modeling of an integrating sphere reflectometer, *Appl. Opt.* 42 (2003) 3832–3842.
- [18] M. Kim, W.D. Philpot, Development of a laboratory spectral backscattering instrument: design and simulation, *Appl. Opt.* 44 (2005) 6952–6961.
- [19] J.M. Davies, W. Zagieboylo, An integrating sphere system for measuring average reflectance and transmittance, *Appl. Opt.* 4 (1965) 167–174.
- [20] L.M. Hanssen, Effects of non-Lambertian surfaces on integrating sphere measurements, *Appl. Opt.* 35 (1996) 3597–3606.
- [21] G. Zaid, S.N. Park, S. Park, D.H. Lee, Differential spectral responsivity measurement of photovoltaic detectors with a light-emitting-diode-based integrating sphere source, *Appl. Opt.* 49 (2010) 6772–6783.
- [22] S. Park, D.H. Lee, S.N. Park, C.W. Park, Experimental validation of the six-port design for a highly uniform integrating sphere photometer, *Appl. Opt.* 52 (2013) 7178–7185.
- [23] S. Park, D.H. Lee, S.N. Park, Six-port integrating sphere photometer with uniform spatial response, *Appl. Opt.* 50 (2011) 2220–2227.
- [24] J. Hwang, H. Cho, D.J. Shin, K.L. Jeong, Correction of port reflection effect for integrating sphere-based reflection measurements, *Metrologia* 50 (2013) 472–481.
- [25] Y. Chen, K.N. Liou, A Monte Carlo method for 3D thermal infrared radiative transfer, *J. Quant. Spectrosc. Radiat. Transfer* 101 (2006) 166–178.
- [26] Huai Chun Zhou, Dong Lin Chen, Qiang Cheng, A new way to calculate radiative intensity and solve radiative transfer equation through using the Monte Carlo method, *J. Quant. Spectrosc. Radiat. Transfer* 83 (2004) 459–481.
- [27] B.X. Li, X.J. Yu, L.H. Liu, Backward Monte Carlo simulation for apparent directional emissivity of non-isothermal semitransparent slab, *J. Quant. Spectrosc. Radiat. Transfer* 91 (2005) 173–179.
- [28] Hannu Parviainen, Karri Muinonen, Bidirectional reflectance of rough particulate media: ray-tracing solution, *J. Quant. Spectrosc. Radiat. Transfer* 110 (2009) 1418–1440.
- [29] Chuan Xiong, Jiancheng Shi, Simulating polarized light scattering in terrestrial snow based on bicontinuous random medium and Monte Carlo ray tracing, *J. Quant. Spectrosc. Radiat. Transfer* 133 (2014) 177–189.
- [30] Siew Kan Wan, Zhixiong Guo, Correlative studies in optical reflectance measurements of cerebral blood oxygenation, *J. Quant. Spectrosc. Radiat. Transfer* 98 (2006) 189–201.
- [31] J.M. Zhao, J.Y. Tan, L.H. Liu, Monte Carlo method for polarized radiative transfer in gradient-index media, *J. Quant. Spectrosc. Radiat. Transfer* 152 (2015) 114–126.
- [32] Jacob C. Jonsson, Geoffrey B. Smith, Gunnar A. Niklasson, Experimental and Monte Carlo analysis of isotropic multiple Mie scattering, *Opt. Commun.* 240 (2004) 9–17.
- [33] C. Calba, C. Roze, T. Girasole, L. Mees, Monte Carlo simulation of the interaction between an ultra-short pulse and a strongly scattering medium: the case of large particles, *Opt. Commun.* 265 (2006) 373–382.
- [34] Z. Gong, H.T. Chen, S.H. Xu, Y.M. Li, L.R. Lou, Monte-Carlo simulation of optical trap stiffness measurement, *Opt. Commun.* 263 (2006) 229–234.